

Two-level evaluation methodology

ARCHITECTURE PATH

Workloads

AlexNet CONV1
AlexNet-derived
dense CONV2

Timeloop

Fixed mappings

Architecture statistics

Action counts
Cycles | utilization

Accelergy + CACTI

Component action
energies

Output

Architecture-level
energy

CIRCUIT PATH

45nm HP.pm PTM model card

Exact committed file

ngspice 42

Transient
simulation

Representative link testbench

CMOS driver -> capacitive link
-> receiver

Circuit outputs

Switching energy/toggle
Propagation delay

Cross-level interpretation

Independent evidence
No co-simulation
Percentages not combined